

Analytic inversion of two-dimensional Fourier optics in aberration-corrected LEEM

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26 August 2026

Abstract

We give the closed-form inverse of discrete bilinear Fourier optics (FO) for a two-dimensional through-focal LEEM series, against the kernel of Yu, Lau and Altman [1]. Intensity at difference frequency ξ is linear in a lag diagonal of the rank-1 Gram $X(q, q') = \Psi(q)\overline{\Psi(q')}$. A vacuum-gauged spectrum is the DC-column factor of X . Subtracting the off-axis remainder reduces each lag to a vacuum 2×2 that Cramer's rule inverts at vanishing Tikhonov parameter whenever the Gram determinant is nonzero. The remainder vanishes on one-, two- and three-mode objects and on non-collinear three-wave objects $\{0, u, v\}$. Under perfect coherence and pure defocus the kernel is a cisoid whose frequency depends on q only through $q \cdot \xi$; equally spaced defoci are a Vandermonde system for the iso- ω fiber masses, which become Gram entries on transversal support. Householder partners across ξ share a node, so they remain one mass. The identities are machine-checked in Lean 4. This is not an inverse of unrestricted Yu $R_{\text{FO}2}$ on a general lattice: the DC slice of a unimodular diagonal kernel has rank one, and radial pure defocus cannot separate perpendicular frequencies.

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1 Purpose and standing hypotheses

The forward map is the FO kernel of Yu, Lau and Altman [1], in the line of Pang *et al.* [3] and Yu *et al.* [2]. The companion inverse note [7] recommends a *linearized* estimator: multi-defocus Tikhonov on the CTF slice $R_{\text{FO}}(\mathbf{q}, 0, \Delta z)$, optionally followed by Gauss–Newton on the bilinear residual. That map is the Fréchet derivative of FO at a strong specular beam. The present note inverts the *bilinear* map itself, in closed form, on a finite abelian frequency group. It is a derivation of identities, not an experimental protocol and not a substitute for [7] on a general micrograph.

Standing discrete data. Let G be a finite abelian group of sampled spatial frequencies (in Lean: any `AddCommGroup` with `Fintype`) and let κ index a finite through-focal series. The object spectrum is $\Psi : G \rightarrow \mathbb{C}$. A kernel at one defocus is a map $R : G \times G \rightarrow \mathbb{C}$; a stack is written R_k . The Yu 2D kernel R_{FO2} is sampled on G by an embedding $\iota : G \rightarrow \mathbb{R}^2$ (`sampledR`). Wave aberration χ is in waves, $W = e^{2\pi i \chi}$, with $\mathbf{q} = \boldsymbol{\alpha}/\lambda$ after acceleration [5, 4, 6].

The reconstruction proceeds in seven steps (Sections 2 to 7 and 9). Each step is an identity; Lean names are given after the statement. No probability space, sampling theorem, or Gauss–Newton convergence estimate is used.

2 Step 1: lift intensity to a Gram contraction

The discrete Fourier coefficient of recorded intensity at difference frequency $\xi \in G$ is

$$\hat{I}(\xi) = \sum_{q \in G} \Psi(q) R(q, q - \xi) \overline{\Psi(q - \xi)}. \quad (1)$$

Define the rank-1 Gram

$$X(q, q') := \Psi(q) \overline{\Psi(q')}. \quad (2)$$

Reassociating (1) yields the linear contraction

$$\hat{I}(\xi) = \sum_{q \in G} R(q, q - \xi) X(q, q - \xi). \quad (3)$$

Thus each fixed- ξ through-focal series is a linear image of the lag diagonal $q \mapsto X(q, q - \xi)$. Inverting bilinear FO splits into (i) inverting those linear slices for X and (ii) factoring the Gram for Ψ .

Lean: `ihat`, `gram`, `ihat_eq_gram`.

Lemma 2.1 (Hermitian intensity). *If $\overline{R(q, q')} = R(q', q)$ for all q, q' , then $\overline{\hat{I}(\xi)} = \hat{I}(-\xi)$.*

Lean: `ihat_hermitian`, and `R_F02_hermitian` for the Yu kernel (nonnegative energy spread).

Lemma 2.2 (Rank-1 pupil). *If $R(q, q') = P(q) \overline{P(q')}$, then $\hat{I}$ is the Gram autocorrelation of $q \mapsto \Psi(q)P(q)$.*

Lean: `ihat_of_rank1`. Perfect coherence of R_{FO2} is of this form (`R_F02_eq_rank1_of_perfect`, `ihat_sampledR_perfect`).

3 Step 2: quotient the global phase

Intensity is invariant under $\Psi \mapsto e^{i\theta}\Psi$:

$$X(e^{i\theta}\Psi) = X(\Psi), \quad \hat{I}(R, e^{i\theta}\Psi) = \hat{I}(R, \Psi). \quad (4)$$

Lean: `gram_eq_of_phase`, `ihat_gauge`.

The inverse therefore recovers Ψ only in a gauge. The vacuum gauge takes the specular coefficient real and nonnegative,

$$\hat{\Psi}(q) = \frac{X(q, 0)}{\sqrt{\|X(0, 0)\|}}, \quad (5)$$

which uses only the DC column of X .

Theorem 3.1 (Vacuum-gauge recovery). *If $\Psi(0) \neq 0$, $\text{Im } \Psi(0) = 0$ and $\text{Re } \Psi(0) \geq 0$, then (5) applied to $X(\Psi)$ returns Ψ .*

Lean: `vacuumGaugePsi`, `vacuumGaugePsi_recovers`, `analyticPsi_recovers`. Without the phase convention one obtains only a unimodular factor $\overline{\Psi(0)}/|\Psi(0)|$ (`vacuumGaugePsi_gram`).

Corollary 3.2 (Uniqueness from the DC column). *If two vacuum-gauged spectra (each with nonzero real-nonnegative specular coefficient) have the same DC column $q \mapsto X(q, 0)$, they coincide.*

Lean: `analyticPsi_unique_of_dc_column`. A full Gram is stronger than needed (`analyticPsi_unique`).

Consequently it is enough to recover $X(\cdot, 0)$, including the scalar $X(0, 0)$, and then apply (5).

4 Step 3: split each lag into vacuum 2×2 plus remainder

For $\xi \neq 0$ the sum (3) isolates the two vacuum pairs $(\xi, 0)$ and $(0, -\xi)$:

$$\hat{I}(\xi) = R(\xi, 0) X(\xi, 0) + R(0, -\xi) X(0, -\xi) + \mathcal{B}(R, \Psi; \xi), \quad (6)$$

where the bilinear remainder is the leftover sum

$$\mathcal{B}(R, \Psi; \xi) = \sum_{q \notin \{0, \xi\}} R(q, q - \xi) X(q, q - \xi). \quad (7)$$

Lean: `ihat_dc_split`, `bilinearRemainder`, `ihat_sub_remainder`.

Subtracting a known remainder produces the vacuum model

$$\hat{I}(\xi) - \mathcal{B}(R, \Psi; \xi) = h U + g V, \quad (8)$$

with through-focal columns $h_k = R_k(\xi, 0)$, $g_k = R_k(0, -\xi)$ and unknowns $U = X(\xi, 0)$, $V = X(0, -\xi)$.

Write

$$\Delta = \left(\sum_k |h_k|^2 \right) \left(\sum_k |g_k|^2 \right) - \left| \sum_k \overline{h_k} g_k \right|^2 \quad (9)$$

for the Gram determinant of the 2×2 at vanishing Tikhonov parameter $\alpha = 0$.

Theorem 4.1 (Cramer recovery of the vacuum pair). *If $\xi \neq 0$ and $\Delta \neq 0$, then the $\alpha = 0$ Cramer formula applied to the remainder-corrected stack recovers $(X(\xi, 0), X(0, -\xi))$ exactly. Equal remainder-corrected stacks imply equal vacuum pairs.*

Lean: `twoColumn_recovers`, `analyticPair_of_remainder`, `dcPair_unique_of_remainder`, `twoColumn_injective`, `gramDet`.

The linearized stage-1 pair estimator of [7] is the same Cramer formula with $\mathcal{B}$ replaced by zero (`stage1Pair_eq_analyticPair_linearized`). Exact vacuum-column recovery therefore requires either knowledge of $\mathcal{B}$ or a support that forces $\mathcal{B} = 0$ (Section 5).

5 Step 4: invert when the remainder vanishes

Support restrictions make (7) identically zero, so [Theorem 4.1](#) applies to the raw intensities.

5.1 One Fourier mode

If $\Psi(q) = 0$ for all $q \neq 0$, then

$$\hat{I}(\xi) = \begin{cases} R(0,0) X(0,0), & \xi = 0, \\ 0, & \xi \neq 0. \end{cases} \quad (10)$$

Lean: `ihat_of_dc_support`.

5.2 Two Fourier modes

Let $\text{supp } \Psi \subseteq \{0, \xi\}$ with $\xi \neq 0$ and $2\xi \neq 0$ (so $-\xi$ is not a mode). Then $\mathcal{B} = 0$, the conjugate branch $X(0, -\xi)$ vanishes, and

$$\hat{I}(\xi) = R(\xi, 0) X(\xi, 0). \quad (11)$$

If $\sum_k |R_k(\xi, 0)|^2 \neq 0$, scalar Tikhonov at $\alpha = 0$ recovers $X(\xi, 0)$. Lean: `bilinearRemainder_twoMode`, `ihat_twoMode`, `analyticPair_twoMode`.

The two amplitudes satisfy the quadratic $s(T - s) = |Z|^2$ with $T = |\Psi(0)|^2 + |\Psi(\xi)|^2$ and $Z = X(\xi, 0)$. The roots are

$$s_{\pm} = \frac{T \pm \sqrt{T^2 - 4|Z|^2}}{2}; \quad (12)$$

the specular (larger-DC) root is s_+ whenever $|\Psi(\xi)|^2 \leq |\Psi(0)|^2$. Lean: `twoModePlus`, `twoModeMinus`, `twoMode_roots_pair`, `twoMode_specular`. The DC intensity is $R(0,0)X(0,0) + R(\xi,\xi)X(\xi,\xi)$ (`ihat_twoMode_dc`).

5.3 Three collinear modes

Let $\text{supp } \Psi \subseteq \{-\xi, 0, \xi\}$ with $\xi \neq 0$, $2\xi \neq 0$ and $3\xi \neq 0$. Every leftover index in (7) hits a missing mode, so $\mathcal{B} = 0$ and

$$\hat{I}(\xi) = R(\xi, 0) X(\xi, 0) + R(0, -\xi) X(0, -\xi). \quad (13)$$

[Theorem 4.1](#) applies to the raw stack. Lean: `bilinearRemainder_threeMode`, `ihat_threeMode_axis`, `analyticPair_threeMode`.

5.4 Two-axis three-wave object

Let $\text{supp } \Psi \subseteq \{0, u, v\}$ with $u \neq 0$, $v \neq 0$, $u \neq v$, $2u \neq 0$, $2v \neq 0$ and $v \neq -u$. (The group hypotheses are the precise content of “non-collinear” on a general abelian group: $v - u \notin \{0, u, v\}$ and $-u \notin \{0, u, v\}$.) Then at frequency u the remainder and conjugate branch drop,

$$\hat{I}(u) = R(u, 0) X(u, 0). \quad (14)$$

Scalar Tikhonov at $\alpha = 0$ recovers $X(u, 0)$ whenever $\sum_k |R_k(u, 0)|^2 \neq 0$. The swapped hypotheses at v recover $X(v, 0)$. Lean: `bilinearRemainder_twoAxis`, `ihat_twoAxis`, `ihat_twoAxis_v`, `analyticPair_twoAxis`, `analyticPair_twoAxis_v`, `twoAxis_dcCol_unique`, `twoAxis_dcCol_unique_v`.

The DC intensity is the sum of the three diagonal Gram entries (`ihat_twoAxis_dc`).

Together with a determination of $X(0, 0)$ ([Section 8.1](#)) this is the full DC column of a three-wave object, hence Ψ by [Corollary 3.2](#).

6 Step 5: pass to the 2D Yu kernel

The working 2D kernel uses the disk aperture, radial χ_{S2} , and vector tilt $\mathbf{a} = \nabla\chi_S(\mathbf{q}) - \nabla\chi_S(\mathbf{q}')$, so the spatial envelope E_S sees $\mathbf{q} \cdot \mathbf{q}'$ [1, 6]. On the CTF slice

$$R_{\text{FO2}}(\mathbf{q}, 0, \Delta z) = R_{\text{FO}}(\|\mathbf{q}\|, 0, \Delta z); \quad (15)$$

off axis R_{FO2} is not the radial 1D kernel evaluated at $(\|\mathbf{q}\|, \|\mathbf{q}'\|)$. Lean: `R_F02`, `R_F02_eq_R_F0_axis`.

Theorem 6.1 (Pure-defocus cisoid). *Under perfect coherence ($q_{\text{ill}} = \Delta E = 0$) and $C_3 = C_5 = 0$,*

$$R_{\text{FO2}}(\mathbf{q}, \mathbf{q}', \Delta z) = A(\mathbf{q})A(\mathbf{q}') e^{i\omega(\mathbf{q}, \mathbf{q}') \Delta z}, \quad (16)$$

with aperture indicators A and instantaneous frequency

$$\omega(\mathbf{q}, \mathbf{q}') = \pi\lambda(\|\mathbf{q}\|^2 - \|\mathbf{q}'\|^2) = \pi\lambda(2\langle \mathbf{q}, \mathbf{q} - \mathbf{q}' \rangle - \|\mathbf{q} - \mathbf{q}'\|^2). \quad (17)$$

Lean: `R_F02_pureDefocus_perfect`, `defocus0Omega`, `defocus0Omega_eq_inner`.

At fixed lag $\boldsymbol{\xi} = \mathbf{q} - \mathbf{q}'$, (17) depends on $\mathbf{q}$ only through the projection $\langle \mathbf{q}, \boldsymbol{\xi} \rangle$. Frequencies with the same projection (and matching aperture) are indistinguishable.

Definition 6.2 (Householder reflection). For $\boldsymbol{\xi} \neq \mathbf{0}$,

$$\text{refl}_{\boldsymbol{\xi}}(\mathbf{q}) = \frac{2\langle \mathbf{q}, \boldsymbol{\xi} \rangle}{\|\boldsymbol{\xi}\|^2} \boldsymbol{\xi} - \mathbf{q}. \quad (18)$$

The reflection preserves $\langle \cdot, \boldsymbol{\xi} \rangle$, Euclidean norms, $\|\mathbf{a}\|$, and ω . Lean: `reflectLine`, `defocus0Omega_reflect`, `R_F02_pureDefocus_perp`.

Sampling R_{FO2} on G via ι yields $R'_{\Delta z}(q, q') = R_{\text{FO2}}(\iota q, \iota q', \Delta z)$. Bilinear FO on that group is (1) with this kernel (`sampledR`, `ihat_sampledR`).

7 Step 6: invert iso- ω fibers by a Vandermonde system

Under the hypotheses of Theorem 6.1, substitute (16) into (3) at lag $\boldsymbol{\xi}$. Group $q \in G$ by a fiber index $\text{idx} : G \rightarrow \{0, \dots, n-1\}$ constant on level sets of $\omega(\iota q, \iota(q - \boldsymbol{\xi}))$. The aperture-weighted fiber masses are

$$M_j = \sum_{\text{idx}(q)=j} A(\iota q) A(\iota(q - \boldsymbol{\xi})) X(q, q - \boldsymbol{\xi}). \quad (19)$$

The through-focal coefficient is then the cisoid mixture

$$\hat{I}(\boldsymbol{\xi}, \Delta z) = \sum_{j=0}^{n-1} M_j e^{i\omega_j \Delta z}. \quad (20)$$

Lean: `apertureGram`, `cisoidMass`, `ihat_eq_cisoidMass`.

Take n equally spaced defoci $\Delta z_k = \delta k$, $k = 0, \dots, n-1$, and write $z_j = e^{i\omega_j \delta}$. Equation (20) becomes the Vandermonde system

$$\hat{I}(\boldsymbol{\xi}, \delta k) = \sum_j M_j z_j^k. \quad (21)$$

If $j \mapsto z_j$ is injective, the nodes are distinct, the Vandermonde matrix $V_{ij} = z_i^j$ is invertible, and

$$\mathbf{M} = \mathbf{y} V(z)^{-1} \quad (22)$$

recovers the masses from the samples $y_k = \hat{I}(\boldsymbol{\xi}, \delta k)$. Lean: `cisoidNode`, `recoveredMasses`, `cisoidMasses_unique`, `recoveredMasses_eq`, `ihat_eq_vandermondeRow`, `recoveredMasses_eq_cisoidMass`.

Householder partners (18) share a node, so they contribute to a single M_j . The inverse of (21) recovers *fiber masses*, not individual lattice points on an iso- ω line.

8 Step 7: transversal support turns masses into Gram entries

Definition 8.1 (Transversal support). The pair (Ψ, ξ) is transversal for idx if each fiber carries at most one occupied pair: whenever $\text{idx}(q) = \text{idx}(q')$ and $\Psi(q), \Psi(q - \xi), \Psi(q'), \Psi(q' - \xi)$ are all nonzero, one has $q = q'$.

Lean: `transversalSupport`.

Theorem 8.2 (Mass equals Gram). Assume [Theorem 6.1](#), distinct cisoid nodes, transversal support, and $A = 1$ at every occupied frequency. If $\Psi(q) \neq 0$ and $\Psi(q - \xi) \neq 0$, then

$$M_{\text{idx}(q)} = X(q, q - \xi). \quad (23)$$

In particular, if $\Psi(\xi) \neq 0$ and $\Psi(0) \neq 0$, the vacuum column entry is the mass of the fiber containing $(\xi, 0)$:

$$M_{\text{idx}(\xi)} = X(\xi, 0). \quad (24)$$

Equal through-focal stacks then imply equal Gram entries at every commonly occupied pair.

Lean: `cisoidMass_eq_gram_of_transversal`, `recoveredMasses_eq_gram_of_transversal`, `recoveredMasses_eq_gram_vacuum`, `gram_eq_of_transversal_ihat`.

If both Householder partners are occupied, [Definition 8.1](#) fails on that fiber: the mass is the sum of two Gram entries and cannot be split. That is the geometric content of [Theorem 6.1](#) for the inverse.

8.1 Recovering $X(0, 0)$

Off-DC fibers never produce the scalar $X(0, 0)$. If $R(q, q) = 1$ for all q and $\hat{I}(0) \neq 0$, the DC intensity is the total energy $T = \sum_q |\Psi(q)|^2$. Write $a = X(0, 0) = |\Psi(0)|^2$ and $S = \sum_{q \neq 0} |X(q, 0)|^2$. Then $S = a(T - a)$, so a is a root of the same quadratic as in [Section 5.2](#),

$$a(T - a) = S, \quad a_{\pm} = \frac{T \pm \sqrt{T^2 - 4S}}{2}. \quad (25)$$

The specular choice is the larger root when the DC beam dominates. The identity

$$X(0, 0) = \frac{\sum_q \|X(q, 0)\|^2}{\hat{I}(0)} \quad (26)$$

holds for the *full* true DC column (including a itself); it is a consistency check, not an inverse for the missing entry. Lean: `recoveredDC`, `recoveredDC_gram`, `twoMode_specular`.

The DC *slice* $q \mapsto X(q, q)$ of a unimodular diagonal kernel is not injective: it has rank one. Lean: `dcSlice_not_injective`. Off-DC slice injectivity recovers lag diagonals with $\xi \neq 0$ (`offDiagGram_eq_of_sliceInjective`) but does not invert $\xi = 0$.

9 Reconstruction recipe

The seven identities assemble into the following procedure. Input: a through-focal stack on a finite frequency group, an embedding ι , and (as appropriate) support or fiber-index data. Output: the vacuum-gauged spectrum $\hat{\Psi}$ when the DC column is determined, else the fiber masses (or vacuum pairs) that the data identify.

1. **Form** $\hat{I}(\xi, \Delta z_k)$. After alignment, discrete-Fourier transform each plane. Work with (1); modes $\|\iota q\| > q_{\text{ap}}$ are identically invisible.

2. **Fix the gauge.** Seek Ψ with $\Psi(0)$ real and nonnegative (Theorem 3.1). A global phase will never be recovered (4).
3. **Choose the lag inverse** according to what is known.
 - a. *Known remainder or small support.* If $\mathcal{B}$ is known, or if Ψ is supported on $\{0\}$, $\{0, \xi\}$, $\{-\xi, 0, \xi\}$, or $\{0, u, v\}$ as in Section 5, apply Theorem 4.1 or the scalar Tikhonov identity (11)–(14) at $\alpha = 0$, provided the relevant energy Δ or $\sum |h_k|^2$ is nonzero. This recovers the vacuum column entries that the support sees.
 - b. *Pure defocus, perfect coherence, unknown general support.* Assume $C_3 = C_5 = 0$ and Theorem 6.1. Index q by $\omega(\iota q, \iota(q - \xi))$, sample n equally spaced defoci, and right-multiply by the inverse Vandermonde matrix (22). This recovers fiber masses M_j , not a unique q on each iso- ω line.
 - c. *Transversal support inside the aperture.* If in addition Definition 8.1 holds and $A = 1$ on the occupied set, each recovered mass is a Gram entry (23). The vacuum column is (24) at every occupied ξ .
4. **Assemble the DC column.** Collect every recovered off-DC entry $X(\xi, 0)$. If $R(q, q) = 1$, determine $X(0, 0)$ from the quadratic (25) (two-mode: $S = |X(\xi, 0)|^2$; many modes: S is the sum of squared off-DC column entries). Equation (26) then checks consistency of the completed column.
5. **Factor the Gram.** Apply (5). By Corollary 3.2 the vacuum-gauged object is unique among spectra sharing that DC column.

The linearized Schiske/Wiener map of [7] is step 3a with $\mathcal{B} := 0$ and without a support hypothesis: it is first-order consistent at vacuum and biased for strong phase.

10 What this does not invert

The analytic inverse is closed-form under remainder knowledge, vanishing remainder (small-mode or two-axis support), or transversal support with distinct cisoid nodes. It is *not* an inverse of unrestricted 2D Yu R_{FO2} on a general lattice, for two independent reasons already present in the kernel.

1. **DC rank one.** A unimodular diagonal kernel has a rank-1 DC slice (`dcSlice_not_injective`). Matching off-DC lag diagonals do not determine X on the diagonal $\xi = 0$.
2. **Projection degeneracy.** Pure-defocus ω depends on q only through $q \cdot \xi$ (17). Householder partners (18) share a cisoid node and remain a single fiber mass. No choice of equally spaced Δz_k splits them.

Statistical noise models, continuum Fourier transforms versus the array DFT, wall-clock FFT complexity, and local convergence of Gauss–Newton remain informal, as in [7].

11 Machine-checked fragment

The identities of Sections 2 to 8 are encoded in Lean 4 [8, 9] against the FO kernel of [1, 6]. The statements are finite-dimensional. Sound proofs depend only on `propext`, `Classical.choice` and `Quot.sound`; there is no `sorry`. Full identifier lists are in the theorem file `docs/LINEAR_INVERSE.md` (T10–T15).

Step	Lean modules
1–3	LeemF0/Inverse/Gram.lean, Analytic.lean
4	LeemF0/Inverse/SmallMode.lean
5	LeemF0/Forward/Kernel2.lean, Inverse/Degeneracy.lean
6–7	LeemF0/Inverse/Fiber.lean

12 Conclusions

Discrete bilinear FO is a linear map on lag diagonals of a rank-1 Gram, followed by a vacuum-gauge factorization. The vacuum 2×2 inverts in closed form once the off-axis remainder is known or killed by support; a pure-defocus through-focal series inverts iso- ω fiber masses by a Vandermonde system; and transversal support promotes those masses to Gram entries. The same kernel that prevents an unrestricted lattice inverse—a rank-one DC slice and Householder-glued cisoid nodes—is what the identities use. The reconstruction of [Section 9](#) is therefore the analytic 2D FO inverse, checked against Yu *et al.* [1], not a claim that every 2D lattice object is identifiable from a radial defocus series.

References

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